Reader Report: Week 06

Probabilistic Invariance: VI-BIP and the Core Theorems

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Summary

Wu and Blei (2025) place a prior over binary selectors $z \in \{0,1\}^p$, where $z_j = 1$ indexes a feature in the invariant set S^* (Peters et al., 2016). The posterior $p(z \mid \mathcal{D}) \propto p(z) \prod_{e,i} \hat{g}(Y_i^e \mid X_{ei}^z)/\hat{p}_e(Y_i^e \mid X_{ei}^z)$ rewards selectors for which the pooled conditional $\hat{g}$ dominates each environment-specific conditional $\hat{p}_e$. Consistency (Thm 1) requires $\mu_{\min} > 0$, the minimum KL divergence between $\hat{g}$ and any $\hat{p}_e$ over non-invariant features (Wu and Blei, 2025, Def. 1). Contraction (Thm 2) gives $\mathrm{TV}(p(z \mid \mathcal{D}), \delta_{z^*}) = O(2^p \, e^{-\kappa n E \mu_{\min}})$. Both bounds assume squared loss under a linear model, and the product likelihood weights all environments symmetrically. VI-BIP (Wu and Blei, 2025, Sec. 4) approximates the posterior via mean-field variational inference (Blei et al., 2017), optimizing the ELBO with the U2G gradient estimator.

Critique

Peters et al. (2016), Fan et al. (2024), and Wu and Blei (2025) all enforce the same invariance condition, yet what separates them is what the practitioner assumes about the environments before seeing data.

ICP assumes the least: environments are exchangeable, and adding more can only shrink the accepted set (Peters et al., 2016, Thm. 1). BIP inherits this agnosticism by weighting environments symmetrically in the product likelihood, but enumerates 2^p models to do so. Rearranging Thm 2 gives $nE \gtrsim O(p/\mu_{\min})$, the same functional dependence on environment count and dimensionality as ICP's power requirement.

NegDRO breaks this symmetry. Its minimax problem permits negative environment weights (Fan et al., 2024, Def. 1), penalizing the optimizer for performing well on a specific environment and enforcing risk parity across all pairs. NegDRO can therefore amplify a single harsh intervention when one environment is known *a priori* to be dominant. Neither paper acknowledges this as a structural incompatibility in their assumptions.

The benchmarking evidence inherits this bias. Wu and Blei (2025, Sec. 5) validate VI-BIP on 6,170 yeast genes across CRISPR knockouts where $\mu_{\min} > 0$ holds by construction, but provide no evaluation in observational or low-environment settings where $\mu_{\min}$ is unknown. The theoretical guarantee holds regardless, but the empirical case for VI-BIP in the low-knowledge regime it targets remains unmade.

Questions

Both questions below concern whether the practitioner-knowledge gap between BIP and NegDRO can be bridged.

1. The critique above treats practitioner knowledge as a binary: either environments are exchangeable (ICP/BIP) or one dominant intervention is known (NegDRO). Can this be formalized as a continuous parameter, so that a single framework interpolates between symmetric and asymmetric environment weighting? If so, does the minimax estimation rate vary smoothly along this axis, or is there a phase transition at which asymmetric weighting becomes strictly preferable?
2. Nonparametric extensions of ICP exist for the symmetric case (Heinze-Deml et al., 2018), but no nonparametric analogue of NegDRO's asymmetric weighting has been proposed. Does the practitioner-knowledge distinction identified above survive when the loss is no longer quadratic, or does the symmetry-versus-asymmetry tradeoff collapse under nonparametric conditionals?

References

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